

Khwaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Final Examination: Spring – 2021

Program: B.Sc. in CSE (Batch – 12th) 1st Year 1st Semester

Course Title: Physics - I

Course Code: PHY 1101

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

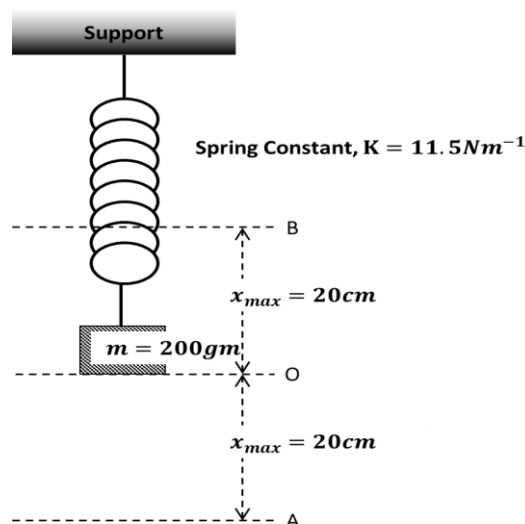
- Which is the unit of mutual inductance?
 - Volt*
 - H
 - Am^{-2}
 - JC^{-1}
 - CV^{-1}
- What amount of angle should be maintained between the current flow and the magnetic field to produce the hall effect in the conductor?
 - 0
 - 30°
 - 90°
 - 180°
 - 360°
- Current flowing solenoid act as a _____.
 - magnet
 - capacitor
 - transistor
 - battery
 - DC generator
- Power of the secondary coil is _____ the power of the primary coil in a step-up transformer.
 - greater than
 - less than
 - negligible to
 - not related to
 - equal to
- What is the value of the proportionality constant of the faraday's law of electromagnetic induction?
 - 1
 - π
 - 4π
 - 10^{-7}
 - $4\pi \times 10^{-7}$
- The frequency of an alternating current is 50 Hz. How long it will take to reach the peak from the zero?
 - 25 sec
 - 0.5 sec
 - 0.02 sec
 - 0.05 sec
 - 0.005 sec

7. The average value of the alternating current for full cycle, $I_{av} = ?$
[Where, I_0 is the peak value]
- a) I_0
 - b) $I_0/2$
 - c) 0
 - d) $0.637I_0$
 - e) $0.707I_0$
8. What is the unit of the reduced mass?
- a) TmA^{-1}
 - b) A^{-1}
 - c) m
 - d) kg
 - e) kg^2
9. Two bodies of masses m_1 and m_2 are connected by a massless spring and their reduced mass is μ . Which condition is correct in that case?
- a) $m_1 < \mu$
 - b) $m_2 < \mu$
 - c) $m_1 < \mu$ but $m_2 > \mu$
 - d) $m_1 > \mu$ but $m_2 < \mu$
 - e) $m_1 > \mu$ and $m_2 > \mu$
10. Time period of the hour hand of a clock is _____.
- a) 24 hours
 - b) 12 hours
 - c) 1 hour
 - d) 60 sec
 - e) 0 sec

(Answer Question No. 1 and any four from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the figure of a spring pendulum and answer the questions that follows:



- a) Investigate the following quantities:
 - i. Angular Frequency [1]
 - ii. Velocity at Point O [1]
 - iii. Acceleration at Point B and [1]
 - iv. Total Energy. [1]
 - b) Construct a wave equation for the above pendulum. [2]
2. a) Show that the acceleration of a simple harmonic oscillator is inversely proportional to the displacement about the mean position. [3]
- b) If a particle of mass 0.05 kg executing simple harmonic oscillation with frequency 50 Hz and if its average kinetic energy is 0.071 J. Calculate its amplitude. [3]
3. a) What are the factors on which the Lissajous figures depends? [2]
- b) Explain the term “All vibrations are periodic but all periodic motions are not vibrations”. [2]
- c) Write any one event that occurs through resonance and give the argument for it. [2]
4. a) Draw a standing wave diagram for 2λ , where λ is the wavelength. [2]
- b) The phase difference produced by two waves is π . what is the Path difference? [2]
- c) Convert the wave equation $y = A \sin(\omega t - \delta)$ to $y = A \sin \frac{2\pi}{\lambda}(vt - x)$. [2]
5. a) Explain how Lenz’s law is in accordance with the law of conservation of energy. [3]
- b) A solenoid 0.5m long has 2000 turns. The magnetic induction near the centre of the solenoid is 0.08T. What is the current in the solenoid? [3]
6. a) Why is the laminated steel core used in the transformer? [2]
- b) Illustrate the rms, average and peak value of alternating current graphically. [2]
- c) The effective voltage of an ideal alternating current is 220 volts. Find the average value for the half cycle of that current. [2]